

Kritika Pandit

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EDUCATION

Carleton College

2022 - 2026

B.A. Computer Science & Statistics | GPA: 3.70/4.00 | Starr International Scholar

Northfield, MN

Graduated with Distinction in Integrative Exercise and Distinction in Major: Computer Science

Coursework: Machine Learning, Data Structures, Algorithms, Statistical Inference, Applied Regression Analysis, Computer Systems, Advanced Statistical Modeling, Probability

PROJECTS

FinLoop - Self-Improving Financial Document Analysis Agent

2026

- Built a 3-agent evaluation loop using the Claude API - improving answer quality from 44.7 to 96.3 out of 100 in a single optimization cycle
- Designed the evaluation rubric before building the loop, logging every prompt version, score, and cost - enabling full observability into system reliability and quality over time

MarketPulse - Real-Time Market Data Platform

2026

- Built a real-time streaming pipeline using Kafka and TimescaleDB - processing live Coinbase and Kraken trade feeds with exactly-once-in-effect aggregation on top of at-least-once delivery
- Built an automated reconciliation system using Airflow - catching a real production bug where server restarts silently dropped data, and self-healing all affected minutes exactly on re-run

Interpretable ML - Forensic Firearm Examiner Decisions

Jan - May 2026

- Trained XGBoost and Random Forest classifiers using scikit-learn on 6,000+ structured records - achieving 80.8% accuracy after correcting for class imbalance and structural missingness
- Applied SHAP values to identify primary predictors of decision outcomes - translating quantitative findings into a written report for a non-specialist audience

Bootstrap CI Methods for Clustered Observational Data

Jan - May 2026

- Designed a Monte Carlo simulation study comparing four statistical methods across 48 scenarios - demonstrating a commonly trusted method silently undercovers the true answer by 25-35 percentage points
- Ran 200 x 300 resamples x 48 scenarios in parallel using SLURM on HPC - validating simulation design against real data before drawing any conclusions

EXPERIENCE

Retail Data Intern - AgenScience AI, Remote

Nov 2025 - Present

- Designed agentic pricing workflows using the Claude API with structured function-calling - translating business stakeholder needs into scalable AI-powered production solutions
- Built Python backend services and REST API integrations on Azure - connecting Shopify to a centralized platform with schema validation at each pipeline stage

Software Engineer Intern - United Nations DESA, Remote

Aug - Oct 2025

- Built Python ETL pipelines using GitHub Actions CI/CD - replacing a 3-day manual process with a fully automated pipeline ingesting 180+ country-level datasets into PostgreSQL
- Communicated technical tradeoffs clearly to non-technical policy stakeholders - keeping decision-makers aligned on data quality risks across a globally distributed team

Student Software Developer - Academic Technology, Carleton College

Jun 2023 - Jun 2026

- Deployed a Llama 3 RAG pipeline using FAISS vector indexing and a Flask REST API - serving 2,000+ users and iterating on retrieval architecture through continuous evaluation
- Built TensorFlow ML pipelines for Drosophila behavioral video analysis - applying feature extraction and classification methods to high-throughput sensor data on GPU hardware

Voice AI Acquisitions Agent

Sep - Nov 2025

- Built a production agentic phone system using LLM function-calling - classifying caller intent and triggering structured downstream actions under a live client deadline
- Designed scalable, well-documented internal APIs connecting Twilio Voice to a PostgreSQL backend - building a React dashboard for real-time monitoring of agent performance

SKILLS

AI & ML: Python, PyTorch, TensorFlow, scikit-learn, XGBoost, RAG pipelines, FAISS, embeddings, retrieval, LLM applications, agentic workflows, prompt engineering, evaluation frameworks

Infra & Practices: SQL, REST APIs, PostgreSQL, Kafka, TimescaleDB, AWS (basic), Azure, Docker, GitHub Actions CI/CD, Git, monitoring, observability, technical documentation